

Current Hybrid Architecture Performance - Fifth Set

Version 10 | Live Vision Hybrid Benchmark | Fifth Set

MODEL_VERSION: 10.11.23

GATE: P2610E2_FIFTH_SET_FULL_POPULATION_LIVE_VISION_HYBRID_ACCURACY_BENCHMARK_V1_0

DECISION: PASS_WITH_LIMITATIONS

LIVE_COMPLETION: COMPLETE_LIVE_BENCHMARK

MODE: LIVE_BENCHMARK

Hybrid semantic authority: Claude Vision preferred after validation.

Deterministic engineering authority: geometry, cut length, development length, spacers, stirrup engineering, steel calculation.

This report is not a production-readiness claim.

No previous-vs-current comparison is included.

EXECUTION PROVENANCE

HYBRID beams: 143 (100.00%)

FALLBACK beams: 0 (0.00%)

VISION_REUSED: None

VISION_RETRIED_AFTER_HISTORICAL_FAILURE: 8

VISION_NEW_LIVE_CALL: 135

VISION COVERAGE

READY: 143 LIMITED: 0

NOT_READY: 0 Eligible: 143

Blocked: 0 Source available: 143

Attempted: 143 API success: 143 API failed: 0

Schema valid: 143 Semantic usable: 143

HYBRID POPULATION

Beam identification: 98.62% (143 / 145)

Bar identification: 51.81% (572 / 1104)

Correct of detected: 36.89% (211 / 572)

Diameter identification: 76.75%

Steel accuracy: 67.82%

Overall accuracy: 63.78%

Model kg: 27977.137 Bench kg: 41250.823

Signed kg error: -13273.685 Abs kg error: 13273.685

FALLBACK POPULATION

Not applicable / insufficient comparison evidence.

FULL POPULATION

Beam identification: 76.47% (143 / 187)

Bar identification: 39.29% (572 / 1456)

Diameter identification: 76.75%

Steel accuracy: 47.58%

Overall accuracy: 50.06%

Model kg: 27977.137 Bench kg: 58796.332

Signed kg error: -30819.195 Abs kg error: 30819.195

VISION FAILURE ANALYSIS

Counts: {"API_FAILED": 0, "SCHEMA_FAILED": 0, "SEMANTIC_UNUSABLE": 0, "TARGET_NOT_IDENTIFIED": 0, "VISUAL_M

Historical API recovered: 8

COST / EXECUTION

Retries: 0 Reused: 0

Input tokens: 420680 Output tokens: 86796

Runtime s: 1617.698

FORMULAS (QA.2A / QA.3.0)

Beam ID = $\text{detected_gt_beams} / \text{total_gt_beams} * 100$

Bar ID = $\text{identified_gt_bars} / \text{total_gt_bars} * 100$

Correct of detected = $\text{MATCH} / \text{detected_bars} * 100$

Steel = $\max(0, 100 - \text{abs}(\text{hybrid-bench})/\text{bench} * 100)$

Overall = mean(Beam ID, Bar ID, correct of detected, steel)

Diameter is reported separately and is not in the overall mean.

Subset scores use matching estimator+model IDs only.

Ground truth: ESTIMATOR_EXCEL. Workbook mapping may not perfectly represent physical drawing interpretation.

LIMITATIONS

- ESTIMATOR_WORKBOOK_MAPPING_LIMITATION
- STIRRUP_ENGINEERING_UNAVAILABLE
- AMBIGUOUS_WITHHELD

CONCLUSION

HYBRID coverage is 100.0% of executed model beams. Subset overall scores are not both applicable; full-population overall is

SCOPE

Fifth Set only. No Second/Third/Fourth/Sixth comparison.

No historical improvement claim. No production promotion.